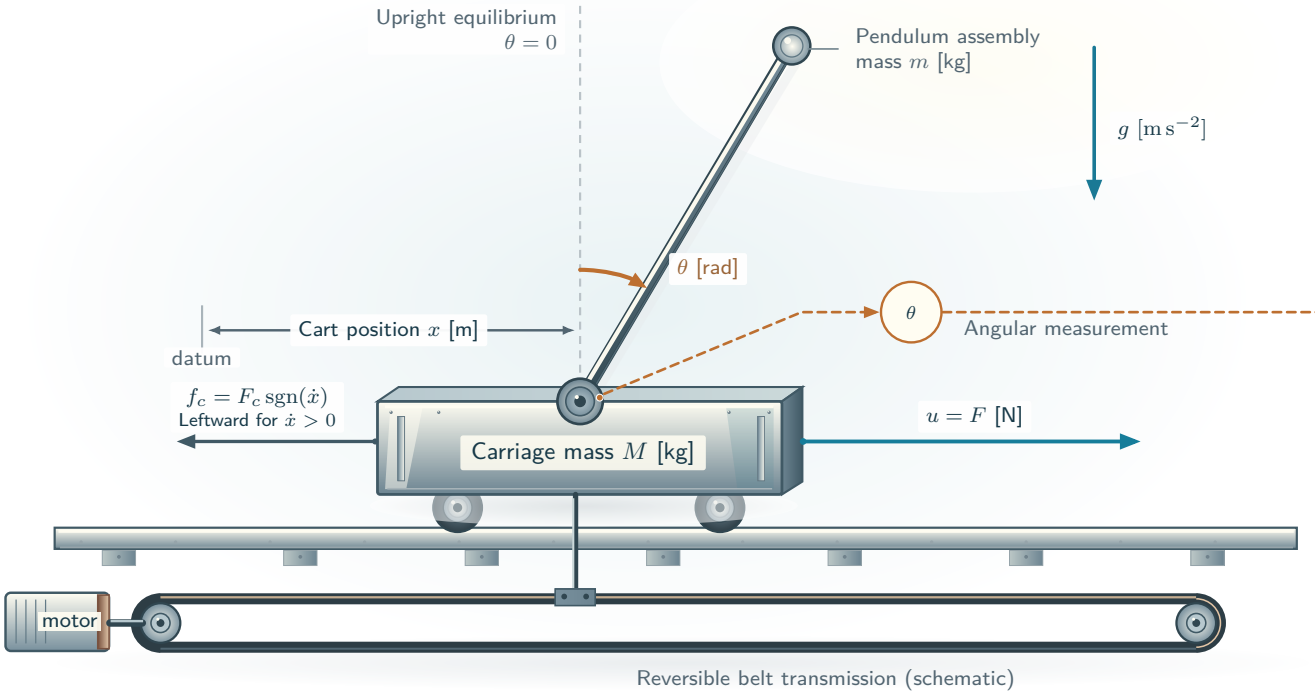
Cart and inverted pendulum

Paper section 3.3: *Cart with inverted pendulum*

Belt-driven carriage and freely rotating pendulum



Reversible belt transmission (schematic)

Control problem

Swing up from the hanging position, balance near the unstable upright state, then position the cart.

Published measurement

Angular position $y = \theta$.

Simulation state order

$[\theta,\ \dot{\theta},\ x,\ \dot{x}]^{\mathsf{T}}$

Units: rad, rad/s, m, m/s.

Nonlinear couplings

Gravity $\sin\theta$, inertial terms $\dot{\theta}^2$, trigonometric coupling and Coulomb friction.

Angular joint loss: $d_{Mf}\dot{\theta}$ [N m].

Input and actuator limit

$u = F$ [N], $|F| \le F_{\max}$.

The source uses a motor and belt; Eq. (32) takes the resulting force as its input.

Length convention

l [m] is the characteristic length used in Eq. (32). The drawing does not infer a separate rod length, inertia or centre of mass from l.

Reading the drawing

Solid arrows: forces / motion.

Dashed arrows: measurement signals.

Geometry is not to scale.

Model scope: Eq. (32) with explicit force saturation and slip friction. No rail-impact or static-sticking model is implied. The educational learning study may declare different friction settings in its case sheet.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.